



Short communication

Multi-view dynastic classification and visual interpretation of blue-and-white porcelain

Edward K.Y. Yapp^{a,1}, Weiming Zhuang^{b,1}, Chen Chen^c, Xianfeng Wang^c, Yonggang Wen^c, Hui-Yuan Yeh^{c,*}^a Singapore University of Social Sciences, 463 Clementi Road, 599494, Singapore^b Sony AI, 1 Venture Avenue, 608521, Singapore^c Nanyang Technological University, 50 Nanyang Avenue, 639798, Singapore

ARTICLE INFO

Handling Editor Nabil Ahmad Afifi

Keywords:

Blue-and-white porcelain

Dataset

Image classification

Visual explanation

Multi-view

ABSTRACT

This study applies deep learning to classify—and interpret—blue-and-white porcelain images by dynastic period and studies the role of multiple object views. A multi-view dataset of Ming and Qing porcelain captured from different angles was curated, resulting in 284 objects and 963 images. Among the evaluated models, a pretrained and fine-tuned ResNet-50 (ImageNet1K-V2) achieved the best performance. Incorporating multiple views of the same object during training improved classification performance compared with a single view. Visual explanation techniques (Grad-CAM, Ablation-CAM, Score-CAM) highlighted shape and motifs as characteristic features. t-SNE visualisation showed learned features cluster by dynasty globally and object locally.

1. Introduction

Advances in Artificial Intelligence (AI) have increasingly found applications in cultural heritage and archaeological studies. Recent works have demonstrated the utility of generative AI for creating complex archaeological illustrations (Magnani & Clindaniel, 2023), metric-learning-based deep convolutional neural networks for site and period prediction (Resler et al., 2021), semantic segmentation models for detection of archaeological sites (Casini et al., 2023; Fuentes-Carbajal et al., 2023) and AI-driven tools for the analysis of numerical and categorical archaeological data (Labba et al., 2023).

Blue-and-white porcelain, exemplified by Jingdezhen's innovations and global reach, embodies the intersection of local craftsmanship, imperial policies, and global trade networks, transforming it into a symbol of Chinese artistry and a key commodity in global cultural exchange (Gerritsen, 2020). Given this historical and cultural importance, traditional archaeological dating and art-historical connoisseurship of Chinese ceramics rely on a systematic analysis of vessel shape (xingzhi), decorative motifs (wenshi), reign marks (kuanshi), glaze characteristics (youze) and body/firing technology (taozhi) (Geng, 1993). However, this process is time consuming and labour intensive.

This complexity makes blue-and-white porcelain a compelling subject for technological exploration, where deep learning methods have emerged as powerful tools to analyse and classify these artefacts. A convolutional neural network (VGG16) was used to classify Chinese Neolithic colour-painted pottery by archaeological culture (Zhao et al., 2023b), while Feature Fusion Ceramic Network (FFCNet) (Liu et al., 2022) and Lightweight Balanced Capsule Network (LBCapsNet) (Li et al., 2024b) were developed to classify different types of blue-and-white porcelain.

Some interesting applications include the use of hyperspectral imaging features for age classification using random forest and Long Short-Term Memory (LSTM) models (Zhao et al., 2023a), graph anomaly detection methods for chemical analysis (Li et al., 2024a), and segmentation of patterns on Yuan blue-and-white porcelain using multi-scale Retinex (a digital image enhancement technique) and histogram multi-peak thresholding (Zheng et al., 2024). These technologies provide deeper insights into material characteristics. More recently, spatial information extracted from VGG16 and ResNet-50 was combined with spectral information to predict the time period of blue-and-white porcelain (Zhao et al., 2025). Object detection using YOLOv11 was applied to classify craftsmanship style, object type and decorative patterns across a

* Corresponding author.

Email address: hyyeh@ntu.edu.sg (H.-Y. Yeh).¹ Joint first author.

range of ceramics; features were then extracted to predict broad market value (Hu et al., 2025). Across these studies, most considered only a single viewpoint of the ceramic artefact. While multiple images of an object from different angles were considered by Ling et al. (2026), the study did not consider how the availability of different views during training affects model performance.

In this paper we investigate whether incorporating multiple views of the same object during training improves classification performance compared with using a single primary view. A curated multi-view dataset comprising 284 objects and 963 images, along with associated metadata, was constructed for Ming and Qing porcelain. Building upon prior work, this study examines the classification performance of blue-and-white porcelain artefacts by dynastic period using a pretrained deep learning framework across multiple evaluation metrics. Beyond classification, visual explanation techniques—Grad-CAM (Selvaraju et al., 2017), Ablation-CAM (Desai & Ramaswamy, 2020) and Score-CAM (Wang et al., 2020)—are employed to qualitatively identify stylistic features influencing model predictions. Furthermore, t-SNE-based feature space visualisation (Van Der Maaten & Hinton, 2008) is used to examine how learned representations cluster by dynastic period and by object. This work contributes to the field by:

- Introducing a curated multi-view dataset of blue-and-white porcelain images and metadata which allows for benchmarking.
- Investigating the role of multi-view training in ceramic classification;
- Combining visual explanation and t-SNE techniques to interpret learned stylistic patterns.
- Open-sourcing the experimental code which promotes reproducibility and fosters collaboration within the scientific community.

2. Background

ImageNet is a large-scale dataset with over 14 million images across 20,000 categories (Deng et al., 2009). It provides a robust foundation for pretraining models, enabling them to learn generalisable features. One such model is Residual Network (ResNet) (He et al., 2016) which is a Convolutional Neural Network (CNN) architecture designed to address vanishing gradient issues in training deep networks. It achieves this through the incorporation of shortcut connections, which enable the network to bypass one or more layers by adding the input of a layer directly to its output. ResNet models are structured as sequences of residual blocks, with each block including a set of convolutional layers and a shortcut connection giving rise to variants such as ResNet-18, ResNet-34 and ResNet-50 (where the suffix corresponds to the number of layers) which differ in their depth and complexity.

Visual explanation methods aim to interpret deep learning model predictions by identifying features that influence decision-making. Grad-CAM (Selvaraju et al., 2017), Ablation-CAM (Wang et al., 2020) and Score-CAM (Desai & Ramaswamy, 2020) are commonly used techniques for generating such explanations in CNNs. Grad-CAM computes the average of the gradients of the target class score with respect to the feature maps in the last convolutional layer, weights the feature maps based on their importance, and then applies a ReLU. Ablation-CAM systematically ablates individual feature maps, measures the drop in class activation score for each map to compute its importance, and combines these weighted activations with ReLU. Score-CAM perturbs the input image with normalised activation maps as masks, measures the resulting change in the target class's score (post-softmax), and combines the maps using these score-based weights.

Dimensionality reduction techniques are essential for analysing and visualising high-dimensional data, particularly in the context of feature representations learned by deep learning models. Principal Component Analysis (PCA) reduces dimensionality by identifying principal components that capture the maximum variance in the data. It transforms the feature space into a lower-dimensional representation while preserving

Table 1
Summary of objects and images by dynasty.

	Number of objects	Number of images
Ming	98	358
Qing	186	605
Total	284	963

as much variability as possible. PCA is computationally efficient and serves as a preprocessing step to reduce complexity before applying more computationally intensive methods.

t-Distributed Stochastic Neighbor Embedding (t-SNE) (Van Der Maaten & Hinton, 2008) is a nonlinear dimensionality reduction technique designed for visualising high-dimensional data in two or three dimensions. It minimises the divergence between probability distributions of data points in the high-dimensional and low-dimensional spaces, ensuring that similar points in the original space remain close in the visualised space. t-SNE is particularly effective for identifying clusters and relationships within data, making it a powerful tool for exploring feature similarities.

3. Dataset

We focus on the blue-and-white porcelain collection from the Metropolitan Museum of Art for two reasons: (1) the museum has an extensive art collection and over 492,000 images of public domain artworks have been made available for free and unrestricted use under its Open Access policy and (2) the dataset is easily accessible through its RESTful API.

Metadata such as Object Number, Is Public Domain, Culture, Period, Object Begin Date, Object End Date, Medium, Country and Classification for 484,957 objects were compiled by the Metropolitan Museum of Art.² First, the dataset was filtered to include only objects for which Is Public Domain was true, Culture contained the word “China” or “Chinese”, Medium contained the word “porcelain”, Country contained the word “China” and Classification contained the word “ceramic”. Second, the object numbers were used to download the primary image for each object as well as any additional images available from different viewpoints; thus, resulting in 3083 objects and 6830 images. Finally, the dataset was manually filtered for blue-and-white porcelain objects and further cleaned to remove images with duplicates or poor lighting. Where the dynasty was not provided under Period, it was inferred from the Object Begin Date: objects were assigned to the Ming dynasty if the date fell within the range [1368, 1644], to the Qing dynasty if it fell within [1644, 1912] and to the Yuan dynasty if it fell within [1271, 1368]; no images were available for the other dynasties. We focus on the Ming and Qing dynasties as there were relatively few images for the Yuan dynasty, reducing the dataset from 299 objects and 986 images to a final dataset of 284 objects and 963 images³. The number of objects and images by dynasty is shown in Table 1.

All objects in the dataset are intact pieces as opposed to shards. Image resolutions range from 267 × 339 to 4000 × 4032 pixels.

4. Methodology

All experiments were conducted on a system equipped with an NVIDIA GeForce RTX 4090 GPU (24 GB memory) and an NVIDIA GeForce RTX 5090 GPU (32 GB memory), running Ubuntu 24.04.1 LTS. All models were implemented using the PyTorch deep learning framework.

ResNet-18, ResNet-34 and ResNet-50 architectures were evaluated. ResNet-18 and ResNet-34 were initialised with ImageNet1K-V1 pretrained weights, while ResNet-50 was initialised with either

² <https://github.com/metmuseum/openaccess>

³ <https://github.com/edwardyapp/blue-and-white-porcelain>

ImageNet1K-V1 or ImageNet1K-V2 weights. The V2 variant reflects an updated set of weights with higher accuracy on the ImageNet dataset which consists of 1000 ImageNet categories and about 1.3 million images.

A 70/30 train–test split was applied and the training set was further split evenly into training and validation sets. Stratification was used to account for class imbalance and the dataset was split at the object level—rather than the image level—to prevent images of the same object from leaking across sets.

The weights for the entire network were frozen except for the last fully connected layer so that it can act as a fixed feature extractor. The original fully connected layer was replaced with a classifier head consisting of a dropout layer ($p = 0.5$) followed by a linear layer with two output nodes for binary classification (Ming vs. Qing dynastic period); only the classifier head was fine-tuned on the current dataset. The number of trainable parameters was 1026 for ResNet-18 and ResNet-34, and 4098 for ResNet-50.

Models were trained using a batch size of 256 with the Adam optimiser and a learning rate of 10^{-3} . Cross entropy loss with class weights (computed as the inverse of the number of images per class) was used to account for the slight class imbalance. Early stopping was applied based on validation loss with a patience of 10 epochs and a minimum delta of 0.01; the maximum number of epochs was set to 100.

Training images were resized to 448×448 pixels, augmented through random horizontal flipping and normalised using ImageNet mean and standard deviation values (red, green, blue channels): [0.485, 0.456, 0.406] and [0.229, 0.224, 0.225] respectively. Validation and test sets were resized and normalised in the same manner but without augmentation.

A total of 30 simulation runs were performed and the mean test results along with 95% confidence intervals ($1.96 \times \sigma / \sqrt{n}$ where σ is the standard deviation and n is the number of runs) were reported. For each run a different random seed was used which controls the data shuffling prior to the train–validation split and the initialisation of the classifier head, while a fixed seed was used for the initial train–test split. Model evaluation was performed at the object level using softmax voting; macro-averaged precision, recall and F1 scores were reported to place equal emphasis on both classes.

The best model in terms of validation loss across 30 simulation runs was used for visual explanation and t-SNE analysis of images in the test set. For t-SNE, PCA was first applied to reduce the feature space to 50 components. Subsequently, these features were projected onto two dimensions using t-SNE with a perplexity of 30 and 1000 iterations.

5. Results and discussion

We first investigate whether images can be classified into the Ming and Qing dynastic periods. A residual network with 50 layers (ResNet-50) which has been pretrained on ImageNet1K-V2 is used where the final fully connected layer is fine-tuned on the porcelain dataset. To understand the stability of the results, 30 simulation runs are conducted and the mean and 95% confidence interval of the accuracy, precision and recall are reported in Table 2; this allows for a more reliable comparison than reporting the best performance from a single run. As we can see the model achieves strong classification performance across all metrics and the results are stable.

We next investigate whether network depth influences performance by comparing the results obtained from ResNet-34 and ResNet-18. However ImageNet1K-V2 pretrained weights are not available for these smaller architectures. Therefore, to enable a fair comparison, ImageNet1K-V1 pretrained weights are used for all three models. The performance of the three V1 models is similar; however, a significant improvement is observed when ImageNet1K-V2 weights are used with ResNet-50. The updated training techniques, preprocessing strategies and optimised weights associated with ImageNet1K-V2, which resulted

Table 2

Comparison of performance metrics across 30 simulation runs for different ResNet models for classification by dynastic period. Values are reported as mean $\pm$ 95% confidence interval. V1 and V2 denote the ImageNet1K pretrained weight versions used for model initialisation.

	Accuracy	Precision	Recall
ResNet-50 (V2)	0.8531 $\pm$ 0.0114	0.8455 $\pm$ 0.0153	0.8361 $\pm$ 0.0090
ResNet-50 (V1)	0.7810 $\pm$ 0.0164	0.7668 $\pm$ 0.0152	0.7805 $\pm$ 0.0137
ResNet-34 (V1)	0.7748 $\pm$ 0.0116	0.7561 $\pm$ 0.0122	0.7626 $\pm$ 0.0142
ResNet-18 (V1)	0.8031 $\pm$ 0.0127	0.7870 $\pm$ 0.0138	0.7952 $\pm$ 0.0119

Table 3

Comparison of ResNet-50 (V2) performance metrics across 30 simulation runs using different image selection strategies for each object: all images (All), only the primary image (Primary), primary and bottom images (Primary + Bottom), and all images except the bottom image(s) (All but Bottom). Values are reported as mean $\pm$ 95% confidence interval.

	Accuracy	Precision	Recall
All	0.8531 $\pm$ 0.0114	0.8455 $\pm$ 0.0153	0.8361 $\pm$ 0.0090
All but Bottom	0.7915 $\pm$ 0.0265	0.7864 $\pm$ 0.0327	0.7573 $\pm$ 0.0367
Primary + Bottom	0.8240 $\pm$ 0.0105	0.8108 $\pm$ 0.0125	0.8035 $\pm$ 0.0113
Primary	0.8279 $\pm$ 0.0100	0.8169 $\pm$ 0.0118	0.8003 $\pm$ 0.0116

in improved accuracy on ImageNet, likely contributed to the improved performance in the present classification task.

To assess whether ResNet-50 (V2) is overfitting (or underfitting), the learning curves in terms of the mean loss and accuracy with 95% confidence intervals across the 30 simulation runs are shown in Fig. 1; corresponding curves for the other models may be found in the Supplementary Material S1. Early stopping is triggered when the validation loss plateaus; the validation accuracy does not show much improvement beyond this point. This shows that the model is neither overfitting nor underfitting.

Thus far the models have been trained using all available images. We now investigate the role of multiple views when training the best model, ResNet-50 (V2), and the results are shown in Table 3. For reference All corresponds to using all images and is the same result from Table 2. In many studies, training is performed using only a single primary image of each object and this setup is denoted as Primary. We pay particular attention to bottom images, where available, as they sometimes contain reign marks, although these are not definitive (Davison, 2021). This gives rise to the combinations Primary + Bottom and All but Bottom; All but Bottom includes all other available images, for example, views from the top, side and close-ups.

Training with only the primary image performs reasonably well relative to the All setup which is reassuring. The performance of Primary + Bottom is similar to that of Primary which suggests that bottom images do not add much value in the context of a single view of the object. Surprisingly, All but Bottom performs the worst. As pointed out by Hu et al. (2025), supplementary images are not always consistent and images from the top may not be informative which could have reduced performance. Nevertheless, when all the images are considered together—that is, bottom images combined with contextual information from multiple views—the model achieves the best performance. The high accuracy, precision and recall for the ResNet-50 (V2) model indicate that Ming and Qing porcelain have visually distinct characteristics—vessel shape and decorative motifs—that enable reliable differentiation, and we investigate these in turn.

We investigate how the model distinguishes between Ming and Qing porcelain by examining heat maps generated from three different visual explanation methods as shown in Fig. 2 for two representative Ming (first two columns) and two representative Qing (last two columns) porcelains

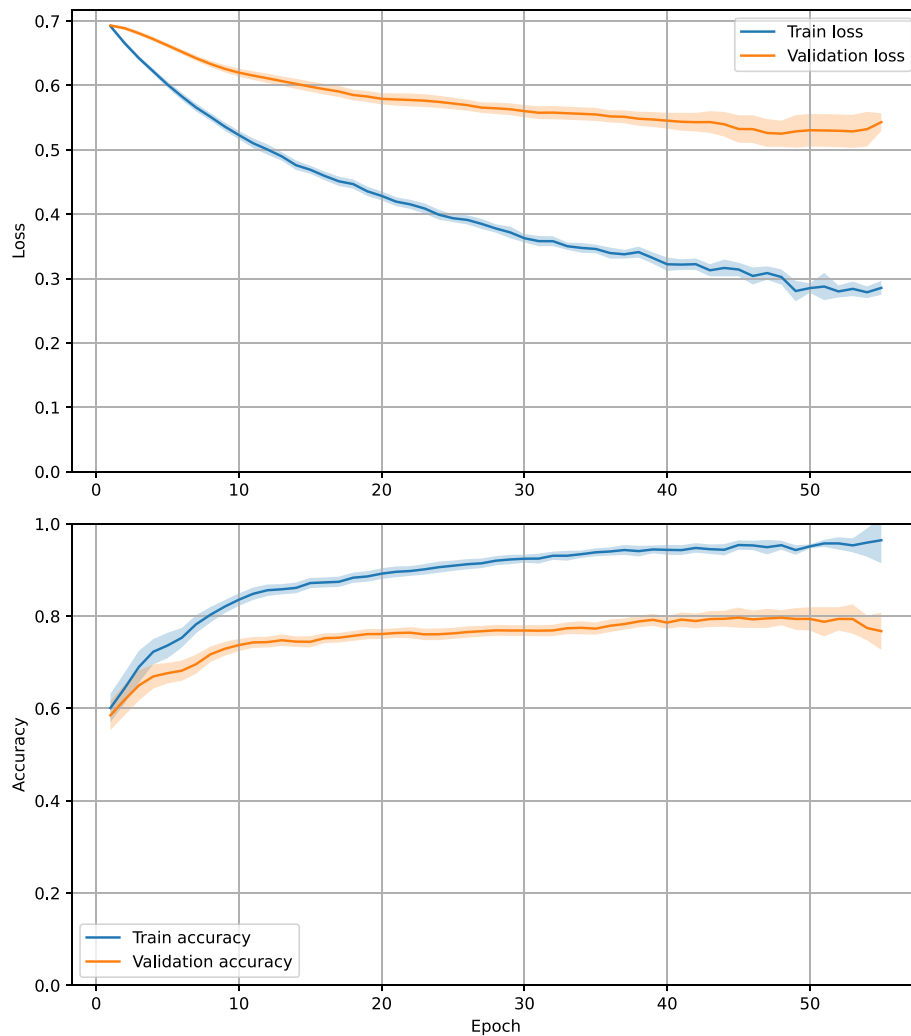


Fig. 1. Average training and validation learning curves across 30 runs for ResNet-50 (V2). The top subplot shows loss and the bottom subplot shows accuracy. The shaded region represents the 95% confidence intervals.

where the red regions correspond to areas contributing most strongly to each predicted category. Note that the results are qualitative as there are no ground-truth labels or pixel-wise annotations specifying which regions definitively characterise Ming or Qing porcelain. Nevertheless, several general observations can be made.

First of all, the heatmaps produced by Grad-CAM, a gradient-based method, and Ablation-CAM, a perturbation-based method, tend to be more similar to each other than to those generated by Score-CAM, which is also perturbation based. This difference highlights the utility of considering multiple visual explanation techniques. Second, consistent activation at the rims and bases, such as the Ming cup in the first column despite the presence of decorative motifs, suggests that the model has learned to prioritise vessel shape (*xingzhi*) and morphology in this instance. In archaeological practice, the profile of the rim and the construction of the foot-rim are critical diagnostic markers for dating. Ming porcelain is traditionally characterised by more robust, clean-lined silhouettes in contrast to the increasingly complex and refined shapes of the Qing dynasty. Third, a recurring theme is the model's focus on narrative iconography, such as the "warrior" figure on the Qing vase in the last column. This alignment with established knowledge is significant, as the transition from the Ming to the Qing saw a shift toward more complex,

secularised storytelling in porcelain decoration (*wenshi*) (Geng, 1993). The model appears to be leveraging these unique iconographic patterns as a proxy for the cultural and temporal shifts recognised by human experts.

The best-performing ResNet-50 (V2) model, selected based on the lowest validation loss across the 30 simulation runs, is used to extract features for t-SNE visualisation of the blue-and-white porcelain test set as shown in Fig. 3. The border colour of each image indicates the dynasty period, while the border style indicates whether the model's prediction is correct or not. Globally, the extracted features form clusters that broadly correspond to the dynastic period, although there is significant overlap. The Ming and Qing dynastic periods spanned hundreds of years including a Transition Period ca. 1620–1629 (Wen-Chin, 1988) during which artistic styles from late Ming continued into the early Qing. Therefore, a degree of stylistic continuity is expected which is consistent with the observed clustering pattern. It is also observed that misclassifications tend to occur near the boundaries between clusters suggesting that these objects may share stylistic features associated with both periods. Locally, multiple images of the same object taken from different angles tend to cluster closely together indicating that the model extracts consistent feature representations across viewpoints.

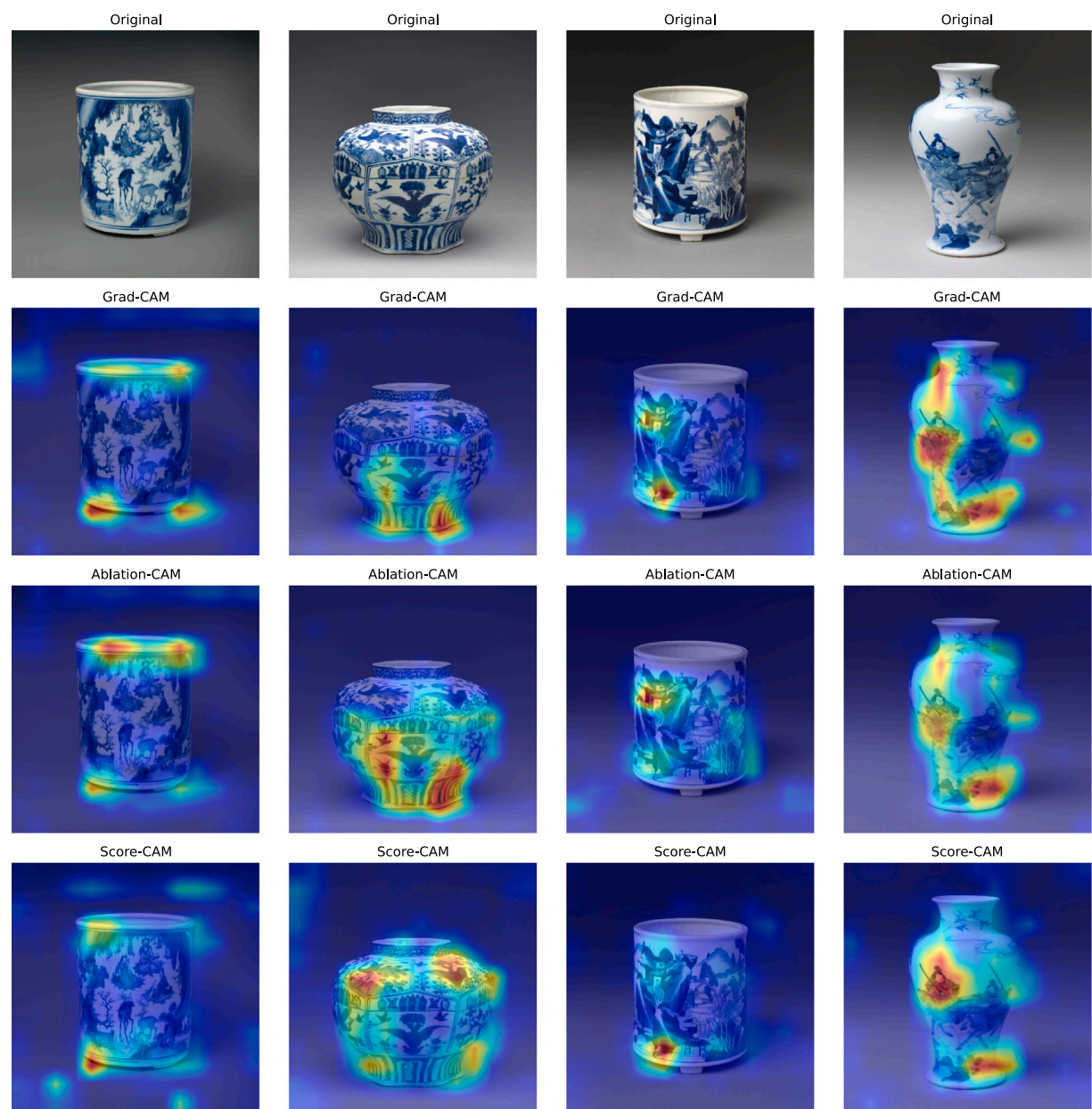


Fig. 2. Original, resized images in the top row and visual explanations using Grad-CAM, Ablation-CAM and Score-CAM in the second, third and bottom rows, respectively. The first two columns correspond to representative Ming porcelain, while the last two columns correspond to Qing porcelain. Red regions correspond to a high score for each respective class.

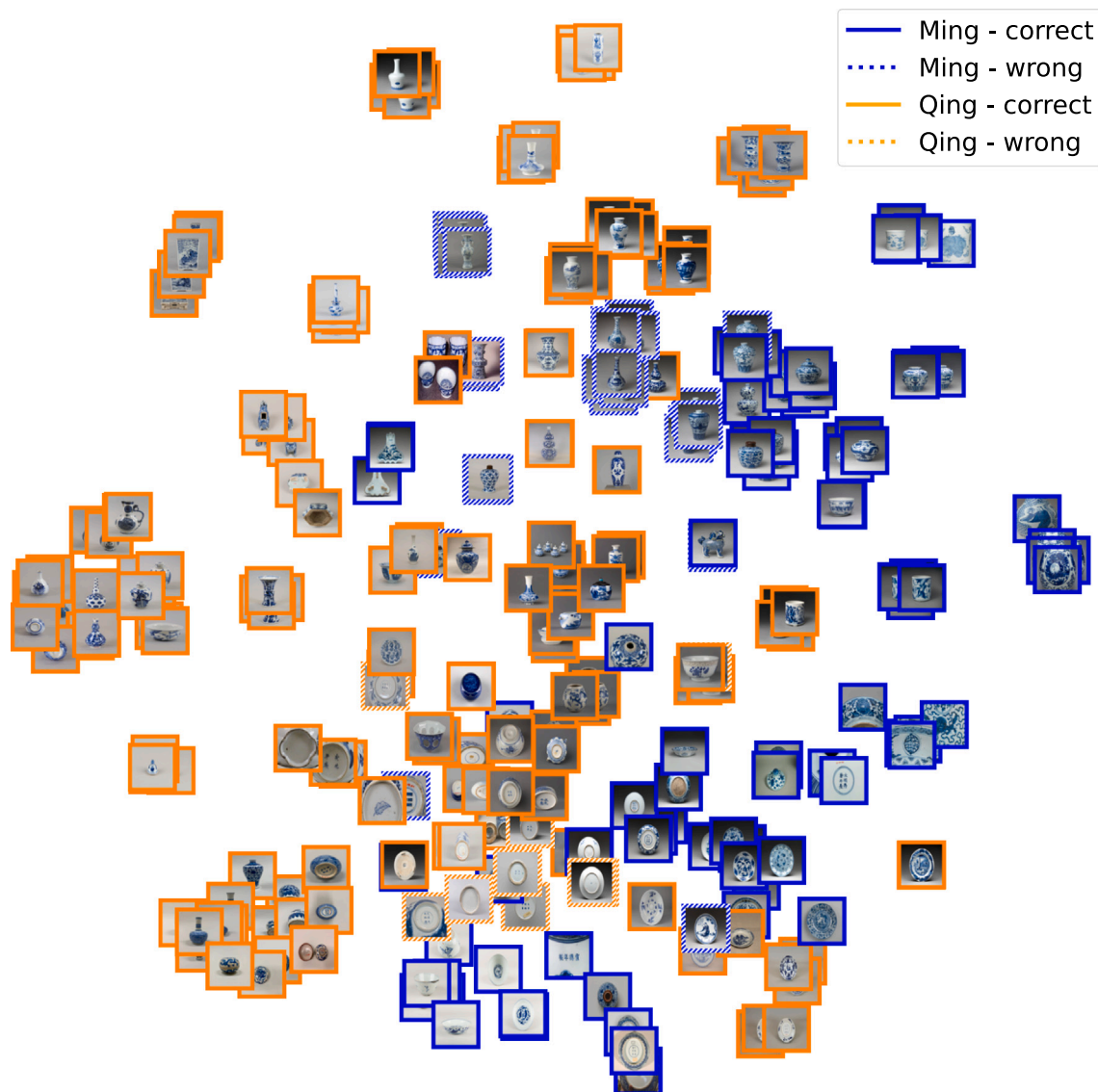


Fig. 3. t-SNE visualisation of test set images from the blue-and-white porcelain dataset. Border colours indicate dynastic period (Ming or Qing), while border style distinguishes correct and wrong model predictions by the ResNet-50 (V2) model.

6. Conclusions

This study demonstrates the potential of deep learning techniques in cultural heritage and archaeology through the classification and analysis of blue-and-white porcelain. A multi-view dataset of Ming and Qing porcelain from the Metropolitan Museum of Art (the Met) was curated, resulting in 284 objects and 963 images with metadata. The Met was selected because of its extensive collection, open access policy and easy API access. Three variations of ResNet models (ResNet-18, ResNet-34, ResNet-50), which were pretrained on ImageNet1K-V1 and V2, were fine-tuned on our dataset and compared. ResNet-50 (V2) achieved the highest accuracy, precision and recall in classifying Ming and Qing porcelain. By varying the training images of each object—from a single primary image to all available views—it was shown that incorporating multiple views of the same object improves classification performance.

The use of multiple visual explanation techniques—Grad-CAM, Ablation-CAM and Score-CAM—indicated that the shape of the object and motifs are key features influencing the model's predictions. t-SNE visualisation of the learned feature space further showed global clustering by dynastic period and local clustering by object.

Future work could expand both the depth and breadth of the dataset to include more images and other dynasties or artefact types besides blue-and-white porcelain by considering open access collections from other museums. Combining the image data with the contextually rich metadata—such as object dimensions and expertly-curated descriptions—would be an interesting direction to further bridge the gap between technology and cultural heritage. Another avenue for future research is to investigate whether there are distinguishing visual features between official kiln blue-and-white porcelain produced for the imperial court and folk kiln blue-and-white porcelain produced for the public or trade.

CRediT authorship contribution statement

Edward K.Y. Yapp: Writing – original draft, Software, Validation. **Weiming Zhuang:** Software, Methodology. **Chen Chen:** Data curation. **Xianfeng Wang:** Writing – review & editing, Conceptualization. **Yonggang Wen:** Writing – review & editing, Conceptualization. **Hui-Yuan Yeh:** Funding acquisition, Conceptualization, Writing – review & editing.

Declaration of generative AI/Generative AI

The authors declare that no AI assisted technologies were used during any stage in the preparation of this article.

Funding statement

This work was funded by Nanyang Technological University, Accelerating Creativity and Excellence (ACE) Award.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.ssaoh.2026.102768.

Data availability statement

Links to code and datasets may be found in the paper.

References

- Casini, L., Marchetti, N., Montanucci, A., Orrù, V., & Rocchetti, M. (2023). A human–AI collaboration workflow for archaeological sites detection. *Scientific Reports*, 13, 8699. <https://doi.org/10.1038/s41598-023-36015-5>
- Davison, G. (2021). *Marks on Chinese ceramics*. Gerald Davison Ltd.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). ImageNet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition* (pp. 248–255). IEEE. <https://doi.org/10.1109/CVPR.2009.5206848>
- Desai, S., & Ramaswamy, H. G. (2020). Ablation-CAM: Visual explanations for deep convolutional network via gradient-free localization. In *2020 IEEE winter conference on applications of computer vision* (pp. 972–980). IEEE. <https://doi.org/10.1109/WACV45572.2020.9093360>
- Fuentes-Carbajal, J. A., Carrasco-Ochoa, J. A., Martínez-Trinidad, J. F., & Flores-López, J. A. (2023). Machine learning and image-processing-based method for the detection of archaeological structures in areas with large amounts of vegetation using satellite images. *Applied Sciences*, 13(11), 6663. <https://doi.org/10.3390/app13116663>
- Geng, B. (1993). *Mingqing Ciqi Jianding [Appraisal of Ming and Qing porcelains]*. Forbidden City Publishing House.
- Gerritsen, A. (2020). *The city of blue and white: Chinese porcelain and the early modern world*. Cambridge University Press. <https://doi.org/10.1017/9781108753104>
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *2016 IEEE conference on computer vision and pattern recognition* (pp. 770–778). IEEE. <https://doi.org/10.1109/CVPR.2016.90>
- Hu, Y., Wu, S., Ma, Z., & Cheng, S. (2025). Integrating deep learning and machine learning for ceramic artifact classification and market value prediction. *Heritage Science*, 13, 306. <https://doi.org/10.1038/s40494-025-01886-6>
- Labba, C., Alcouffe, A., Crubézy, E., & Boyer, A. (2023). IArch: An AI tool for digging deeper into archaeological data. In *2023 IEEE 35th international conference on tools with artificial intelligence* (pp. 22–29). IEEE. <https://doi.org/10.1109/ICTAI59109.2023.00012>
- Li, J., Yang, Y., Zou, D., Wu, L., Wu, Q., Lin, Z., & Li, Q. (2024a). Recognition of Yuan blue and white porcelain produced in Jingdezhen based on graph anomaly detection combining portable X-ray fluorescence spectrometry. *Heritage Science*, 12, 79. <https://doi.org/10.1186/s40494-024-01193-6>
- Li, R., Geng, G., Wang, X., Qin, Y., Liu, Y., Zhou, P., & Zhang, H. (2024b). LBCapsNet: A lightweight balanced capsule framework for image classification of porcelain fragments. *Heritage Science*, 12, 133. <https://doi.org/10.1186/s40494-024-01250-0>
- Ling, Z., Delnevo, G., Salomoni, P., & Mirri, S. (2026). Multi-task learning for identification of porcelain in Song and Yuan dynasties. *Neural Computing and Applications*, 38, 37. <https://doi.org/10.1007/s00521-025-11703-7>
- Liu, Y., Liu, B., Yu, J., Xia, J., & Luo, C. (2022). Automatic classification of blue and white porcelain sherds based on data augmentation and feature fusion. *Applied Artificial Intelligence*, 36(1), 1994232. <https://doi.org/10.1080/08839514.2021.1994232>
- Magnani, M., & Clindaniel, J. (2023). Artificial intelligence and archaeological illustration. *Advances in Archaeological Practice*, 11(4), 452–460. <https://doi.org/10.1017/aap.2023.25>
- Resler, A., Yeshurun, R., Natalio, F., & Giryas, R. (2021). A deep-learning model for predictive archaeology and archaeological community detection. *Humanities and Social Sciences Communications*, 8, 295. <https://doi.org/10.1057/s41599-021-00970-z>
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *2017 IEEE international conference on computer vision* (pp. 618–626). IEEE. <https://doi.org/10.1109/ICCV.2017.74>
- Van Der Maaten, L., & Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9, 2579–2605. <http://www.jmlr.org/papers/volume9/vandemaaten08a/vandemaaten08a.pdf>
- Wang, H., Wang, Z., Du, M., Yang, F., Zhang, Z., Ding, S., Mardziel, P., & Hu, X. (2020). Score-CAM: Score-weighted visual explanations for convolutional neural networks. In *2020 IEEE/CVF conference on computer vision and pattern recognition workshops* (pp. 111–119). IEEE. <https://doi.org/10.1109/CVPRW50498.2020.00020>
- Wen-Chin, H. (1988). Social and economic factors in the Chinese porcelain industry in Jingdezhen during the late Ming and early Qing period, ca. 1620–1683. *Journal of the Royal Asiatic Society of Great Britain & Ireland*, 120(1), 135–159. <https://doi.org/10.1017/S0035869X00164196>
- Zhao, H., Hu, Z., Liu, G., Xu, S., Lu, Z., & Zheng, Q. (2023a). Research on blue and white porcelain from different ages based on hyperspectral technology. *Journal of Cultural Heritage*, 62, 151–159. <https://doi.org/10.1016/j.culher.2023.05.025>
- Zhao, H., Tang, G., Hu, Z., Xie, Y., Liu, G., Lu, Z., & Wang, P. (2025). Dating of Jingdezhen blue and white porcelain based on transfer learning and imaging spectroscopy techniques. *Heritage Science*, 13, 68. <https://doi.org/10.1038/s40494-025-01652-8>
- Zhao, X., Shu, C., Jiang, S., & Hu, Y. (2023b). From classification to matching: A CNN-based approach for retrieving painted pottery images. *Digital Applications in Archaeology and Cultural Heritage*, 29, e00269. <https://doi.org/10.1016/j.daach.2023.e00269>
- Zheng, Q., Zhu, B., Cai, Q., Li, J., Fang, C., & Wu, N. (2024). Extraction method of Yuan blue and white porcelain pattern based on multi-scale Retinex and histogram multi-peak threshold segmentation. *Heritage Science*, 12, 211. <https://doi.org/10.1186/s40494-024-01324-z>